

Project Proposal Form

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Centre Name	<u>British College of Applied Studies</u>	Centre Number	<u></u>
Assessor	<u>Mrs. ALF. Sajeetha</u>	Date	<u>2025.02.21</u>
Unit	<u>13: Computing Research Project</u>	Batch No	<u>CSD - 22</u>

Proposed project title Evaluating the Performance of Object Detection Models for Accurate Sri Lankan Traffic Sign Detection

Section One: Title, aims, objectives

Title or working title of project (in the form of a question, objective or hypothesis)

Research Question:

- Which object detection model (YOLOv8, Faster R-CNN or SSD) achieves the highest accuracy in detecting Sri Lankan traffic signs, as measured by mean Average Precision (mAP) and F1-Score?
- What is the most accurate object detection model among YOLOv8, Faster R-CNN, and SSD for detecting Sri Lankan traffic signs, considering mean Average Precision (mAP) and F1-Score? (revised)

Hypothesis:

- Among the YOLOv8, Faster R-CNN and SSD, one of the object detection models will achieve the highest mAP in detecting Sri Lankan traffic signs, with performance variations due to differences in model architecture and detection capabilities.

Project aims and objectives (e.g. what do you wish to achieve? What do you want to learn to do? What do you want to find out?)

Project aim:

- To compare the performance of YOLOv8, Faster R-CNN, and SSD for Sri Lankan traffic sign detection and identify the model with the highest mAP, contributing to improved automated traffic sign recognition and enhanced road safety in Sri Lanka using AI.

Project objectives:

- To conduct a comprehensive comparison of existing object detection models (YOLOv8, Faster R-CNN and SSD) in terms of their suitability for traffic sign detection in Sri Lanka

contexts.

- To evaluate the performance of YOLOv8, Faster R-CNN and SSD models on a Sri Lankan traffic sign dataset using the mAP metric.
- To compare and analyze the performance of these models, identifying the most effective model for accurate detection of Sri Lankan traffic signs.

Section Two: Reasons for choosing this project

Reasons for choosing the project (e.g. links to other subjects you are studying, personal interest, future plans, knowledge/skills you want to improve, why you feel the topic is important)

- This project was chosen due to my strong personal interest in computer vision and its applications, particularly in the field of object detection.
- As I pursue a career in this domain, focusing on real-world problems like traffic sign detection aligns with my professional goals.
- Moreover, traffic sign detection plays a crucial role in the development of autonomous vehicles, where accurate and reliable perception systems are essential for ensuring road safety and improving vehicle navigation.

Section Three: Precedents and sources

Use of key literature sources and precedents to support your project objectives

Khan, A., & Sohail, S. (2023). Deep Learning Based Autonomous Real-Time Traffic Sign Recognition System for Advanced Driver Assistance. *International Journal of Image Processing and Computer Vision*, 14(6), 1-10. - This paper proposes a real-time traffic sign recognition system using deep learning, employing a two-stage approach: YOLO-based model for traffic sign identification and Xception-based model for classification.

Mishra, A., et al. (2022). Evaluating and Bench-marking Object Detection Models for Traffic Sign and Traffic Light Datasets. *ACCVW 2022*. - This paper evaluates and benchmarks various object detection models on traffic sign and traffic light datasets, analyzing their performance and establishing a benchmark for future research.

Zhao, Z., et al. (2023). Performance Comparison of Object Detection Models for Road Sign Detection Under Different Conditions. *arXiv preprint arXiv:2307.13723*. - This paper evaluates the performance of state-of-the-art object detection models (YOLOv8, YOLOv9,

RTMDet, RetinaNet, Faster R-CNN) for road sign detection under various conditions, comparing their accuracy, speed and loss.

Section Four: Activities and timescales

Activities to be carried out during the project

(Not Required to Fill)

How long will this take:

Milestone one: Interim Report Submission

Target date (set by tutor-assessor): As per the Schedule

Milestone two: Final Submission & Presentation

Target date (set by tutor-assessor): As per the Schedule

Section Five: Resources

What resources will you need for your research project, write up and presentation (eg, libraries, books, journals, equipment, rehearsal space, technology and equipment, venue, physical resources, finance):

Books and Journals:

- Books on object detection and computer vision.
- Research papers on YOLO, Faster R-CNN, SSD and traffic sign detection.

Libraries and Frameworks:

- PyTorch and OpenCV for model evaluation and image processing.
- Matplotlib and Seaborn for visualizing the results.
- NumPy and Pandas for data manipulation and analysis.

Dataset:

- Sri Lankan Traffic Sign Dataset (dataset collected from Roboflow).

Technology and Equipment:

- A computer with adequate processing power.
- Cloud computing resources (Google's Colab for running model evaluations, since local machine doesn't have sufficient resources).

Section Six: Justification of the Relevancy within the theme

Justify your project title in terms of how it related to the research theme and how it related your respective specialization (software / network)

The project 'Evaluating the Performance of Object Detection Models for Accurate Sri Lankan Traffic Sign Detection' is directly aligned with the theme of AI, specifically in the field of computer vision. Object detection, a subset of AI, involves the development and application of algorithms that enable machines to identify and locate objects within digital images. This project explores state-of-the-art object detection models such as YOLOv8, Faster R-CNN and SSD, which are integral to advancing AI capabilities in real-world applications.

Furthermore, this project is highly relevant to my software engineering specialization, as it bridges theoretical AI techniques with practical, real-world challenges. The focus on traffic sign detection ties into automated systems like autonomous vehicles, which rely heavily on AI for environmental perception and decision-making. As a software engineer, working with AI-based models enhances my ability to build and optimize AI-driven applications, particularly in fields that require real-time performance and high accuracy, such as autonomous navigation systems.

Comments and agreement from project supervisor

Is the learner taking this project as part of the Higher National Diploma? Yes/No

If yes, which Diploma are they taking? _____

Comments (optional):

Is project derived from work which has been/will be submitted for another qualification? Yes/No

Which qualification (title and unit)? _____

Comments (optional):

I confirm that the project is not work which has been or will be submitted for another qualification and is appropriate and I am willing to supervise the above project for the above stated learner.

Agreed (Yes / No):

(name)

(date)

Comments and agreement from project proposal Assessor

Comments (optional):

I confirm that the project is appropriate.

Agreed (Yes / No):

(name)

(date)